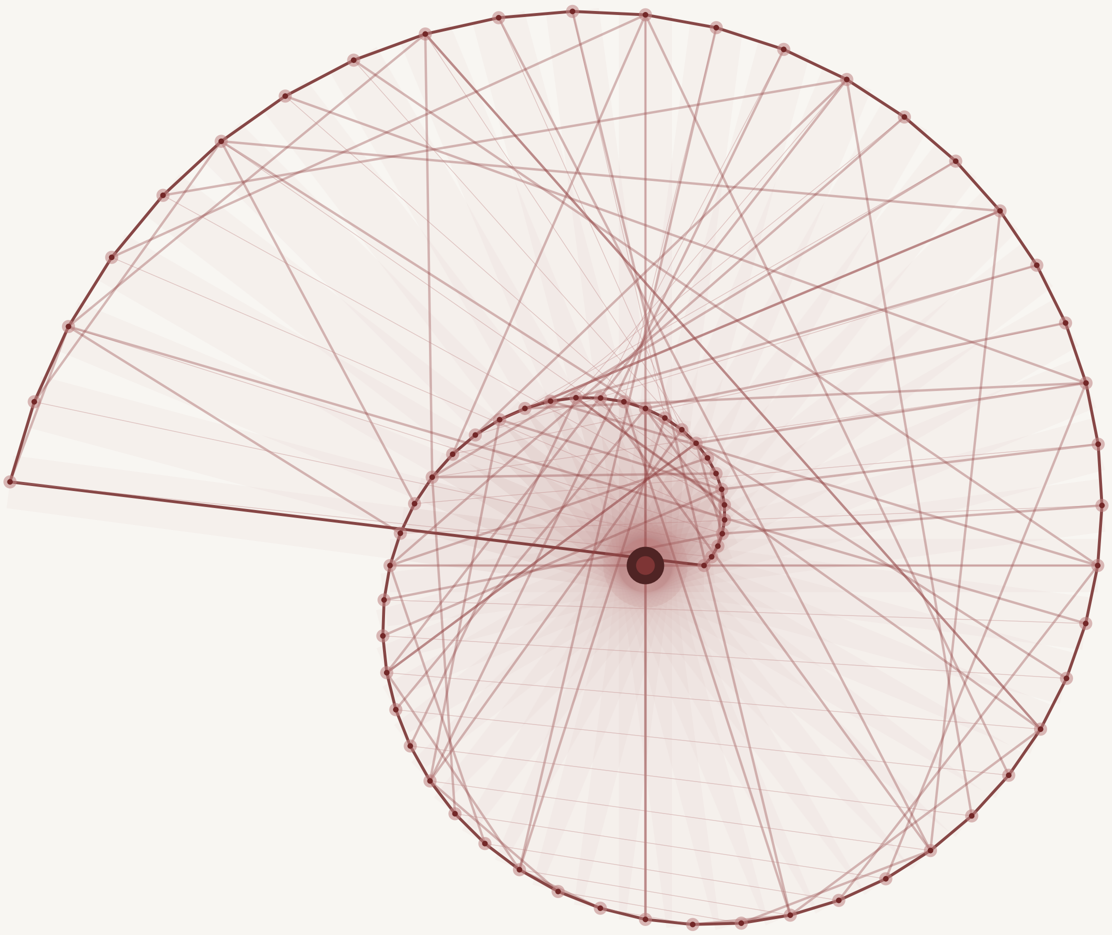
ALGEBRA II

— David Puerta —

Rings ∘ Integral Domains ∘ Ideals and Quotient Rings

Groups ∘ Dihedral and Symmetric Groups ∘ Group Actions



Abstract

These notes are a summary of the Algebra 2 module taught by Alexander Stasinski and Pavel Tumarkin in 2025-26. There will be minimal, if any, proofs as these notes focus on the main results of the course for revision purposes. If you do find any errors or typos, feel free to let me know at davidpuertamaths@gmail.com

Contents

Notation

R or $(R, +, \cdot)$ – a ring

F – a field

$|$ – divides

$\nmid$ – does not divide

$\gcd(\cdot, \cdot)$ – greatest common divisor/highest common factor

$\mathbb{Z}$ – the integers $\{\cdots, -2, -1, 0, 1, 2, \cdots\}$

$\mathbb{N}$ – the positive integers $\{1, 2, 3, \cdots\}$

$R^\times$ – the set of units of a ring R

$\mathbb{Z}/n$ – the integers modulo n

(a) – the principle ideal generated by a

$(a_1, \cdots, a_k)$ – the ideal generated by the elements $a_1, \cdots, a_k$

$\bar{x}$ – the coset of x

R/I – the quotient ring

$\mathbb{F}_{p^n}$ – the finite field with p^n elements

G or $(G, \star)$ – a group

D_n – the dihedral group with $2n$ elements

$\langle a_1, \cdots, a_n \mid A \rangle$ – the group generated by elements $a_1, \cdots, a_n$ given relations A

$|G|$ – order of a group

$\mathrm{ord}(x)$ – the order of the element x

$\mathrm{lcm}(\cdot, \cdot)$ – the lowest common multiple

$\bigsqcup$ – disjoint union

$[G : H]$ – the index of H in G

$H \triangleleft G$ – H a normal subgroup of G

Definitions A–Z

1 Rings

1.1 Definition

A ring R is a set together with two binary operations $+, \cdot$, such that $(R, +)$ is an abelian group and the following hold:

1. $\exists\, 1 \in R$ such that $1 \cdot x = x \cdot 1 = x$ for all $x \in R$.
2. For all $x, y, z \in R$ we have $x(yz) = (xy)z$.
3. For all $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

1.2 The ring $\mathbb{Z}/n$

We have the ring $\mathbb{Z}/n$ which has elements $\bar{a} = \{x \in \mathbb{Z} \mid x \equiv a \pmod{n}\}$.

1.3 Subrings

A subring $S \subset R$ of R is a subset which contains $\{0, 1\}$ and is closed under addition, multiplication and negation.

1.4 Feilds

A field is a ring $(R, +, \cdot)$ where the following hold:

1. R is commutative.
2. $1 \neq 0$.
3. Every non-zero element has an inverse.

These inverses are unique.

2 Arithmetic $\mathbb{Z}$ and $F[x]$

In a ring R, for $a, b \in R$ have that $a \mid b$ if $a = bc$ for some $c \in R$.

2.1 Divisibility in $\mathbb{Z}$

For $a \in \mathbb{Z}$, $b \neq 0$, there exists unique $q, r \in \mathbb{Z}$ such that $a = qb + r$ and $0 \leq r < |b|$.

The Euclidean algorithm for finding $d = \gcd(a, b)$ for some $a, b \in \mathbb{Z}$ is as follows:

1. Write $a = q_1 b + r_1$ for some q_1, r_1.
2. Write $b = q_2 r_1 + r_2$.
3. Write $r_1 = q_3 r_2 + r_3$ and repeat, that is use $r_n = q_{n+2} r_{n+1} + r_{n+2}$ for $n \geq 1$, until $r_n = 0$.
4. Once $r_n = 0$, then $r_{n-1} = d$.

If you wish to write $ax + by = d$ for some $x, y \in \mathbb{Z}$, then rearrange all lines for r_i and substitute them all into the penultimate line, eliminating all except a and b.

For instance,

$$
\begin{aligned}
1235 &= 2 \cdot 455 + 325 \\
455 &= 1 \cdot 325 + 130 \\
325 &= 2 \cdot 130 + 65 \\
130 &= 2 \cdot 65 + 0
\end{aligned}
$$

which gives $\gcd(455, 1235) = 65$. Rearranging the penultimate line for 65 gives

$$65 = 325 - 2(130)$$

Using the second line, we can rearrange for 130,

$$130 = 455 - 1(325)$$

and similarly, in the first line we can rearrange for 325,

$$325 = 1235 - 2(455)$$

which gives,

$$65 = (1235 - 2(455)) - 2(455 - 1(1235 - 2(455))) = -8(455) + 3(1235)$$

If p is a prime, then $p \mid ab \Rightarrow p \mid a$ or $p \mid b$.

The Fundamental Theorem of Arithmetic states that if $n \in \mathbb{N}$ with $n > 1$, then n can be expressed uniquely as a product of primes, that is

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$$

for some p_i's and α_k's is unique.

2.2 Divisibility in $F[x]$

Let F be a field. For a polynomial $f(x) \in F[x]$ written $f(x) = a_0 + a_1 x + \cdots a_n x^n$ with $a_i \in F$, we define the degree of a polynomial as

$$\deg(f) := \begin{cases} \max\{i \mid a_i \neq 0\} & \text{if } f(x) \neq 0 \\ -\infty & \text{if } f(x) = 0 \end{cases}$$

and we have the following properties for $f(x), g(x) \in F[x]$:

1. $\deg(fg) = \deg(f) + \deg(g)$
2. $\deg(f) + \deg(g) \leq \max\{\deg(f), \deg(g)\}$

For $f(x), g(x) \in F[x]$, $g(x) \neq 0$, there exists unique polynomials $q(x), r(x) \in F[x]$ with $\deg(r) < \deg(g)$ such that $f(x) = q(x)g(x) + r(x)$.

2.3 GCDs and The Euclidean Algorithm in $F[x]$

A polynomial $f(x) \in F[x]$ is monic if the leading coefficient is 1.

2.4 Unique Factorisation, Units and Irreducibles

An element in a ring $a \in R$ is a unit if there exists $b \in R$ such that $ab = 1$. The set of units of R is denoted $R^\times$.

An element in a ring $r \in R$ is irreducible if it satisfies:

1. r is not a unit.
2. If $r = ab$, then a or b is a unit.

If $p \in R$ is irreducible, R either $\mathbb{Z}$ or $F[x]$, and $p \mid ab$ then $p \mid a$ or $p \mid b$.

Let R be either $\mathbb{Z}$ or $F[x]$, and $a \in R$ be a non-zero non-unit element. Then a can be factorised uniquely into a product of irreducible elements, up to order of the factors and multiplication by invertible elements.

3 Units, Zero Divisors and Integral Domains

3.1 Units

We have the following properties.

- For F a field, the units of $F[x]$ are precisely $F^\times$.
- An element $\bar{a} \in \mathbb{Z}/n$ is a unit if and only if $\gcd(a, n) = 1$.
- $\mathbb{Z}/p$ is a field if and only if p is prime.

3.2 Zero Divisors and Integral Domains

For a ring R, a zero divisor is an element $a \in R$ such that for some $b \neq 0$, $ab = 0$.

A ring R with at least two elements, and no zero divisors other than 0 is called an integral domain. A key property of integral domains is the following cancellation property,

$$ax = bx \Rightarrow a = b \text{ for } x \neq 0$$

Any ring that is a subring of a field is an integral domain.

We have the following properties.

- The ring $\mathbb{Z}/n$ is an integral domain if and only if n is prime.
- If R is an integral domain then $R[x]$ is an integral domain.

3.3 Prime Elements and Unique Factorisation Domains

In a ring R, an element $x \in R$ is prime if the following holds:

- x is a non-zero non-unit element.
- If $x \mid ab$ for $a, b \in R$ then $x \mid a$ or $x \mid b$.

In an integral domain R, if x is prime then it is irreducible. However, in some rings the converse may not hold, that is not all irreducible elements are prime.

An integral domain R is a unique factorisation domain if every non-zero non-unit element of R can be written as a product of irreducible elements and this product is unique up to order of the factors and multiplication by units.

In a unique factorisation domain, prime elements and irreducible elements coincide.

4 Factorisation of Polynomials

4.1 Irreducible Polynomials in $F[x]$

For $f(x) \in F[x]$ where F is a field and $\deg(f) \leq 4$ we have the following conditions for irreducibility:

- If $\deg(f) = 1$, then $f(x)$ is irreducible in $F[x]$.
- If $\deg(f) = 2, 3$, then $f(x)$ is irreducible in $F[x]$ if and only if it has no roots over F.
- If $\deg(f) = 4$, then $f(x)$ is irreducible in $F[x]$ if and only if it has no roots over F and it is not the product of two quadratic polynomials.

4.2 Irreducible Polynomials in $\mathbb{Q}[x]$ and $\mathbb{Z}[x]$

The Rational Root Test states that if $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$ with $\deg(f) \geq 1$, then if we have a rational root p/q where $p, q \in \mathbb{Z}$ and $\gcd(p, q) = 1$, then $p \mid a_0$ and $q \mid a_n$.

Gauss' Lemma states that if $f(x) \in \mathbb{Z}[x]$ with $\deg(f) \geq 1$ and that the greatest common divisor of all its coefficients is 1, then $f(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.

Let R be a commutative ring, let $x \in R$ be irreducible and $u \in R^\times$, then ux is irreducible.

Eisenstein's criterion states that if $f(x) = a_0 + a_1x + \cdots a_nx^n \in \mathbb{Z}[x]$ and let $p \in \mathbb{Z}$ be a prime such that $p \mid a_0, a_1, \cdots, a_{n-1}$, $p \nmid a_n$ and $p^2 \nmid a_0$ then $f(x)$ is irreducible in $\mathbb{Q}[x]$.

5 Ring Homomorphisms

A ring homomorphism between two rings R and S is a map $f : R \to S$ with the following:

1. $f(1) = 1$.
2. $f(a + b) = f(a) + f(b)$ for all $a, b \in R$.
3. $f(ab) = f(a)f(b)$ for all $a, b \in R$.

For a ring homomorphism $f : R \to S$, we have the properties:

- $f(0) = 0$.
- $f(-a) = -f(a)$ for all $a \in R$.

Two rings R and S are isomorphic if there exists a bijective homomorphism (an isomorphism) between R and S, we write $R \cong S$.

A nice way to show $f : R \to S$ is injective for rings R and S is to show that $\mathrm{Ker}(f) = \{0\}$.

5.1 Direct Products of Rings

Given two rings R and S, we can construct the direct product of R and S, given by $R \times S$. This new ring is defined by the pairs (r, s) for $r \in R$ and $s \in S$ with operations pairwise.

6 Ideals and Quotient Rings

Let $f : R \to S$ be a homomorphism, then $\text{Im}(f)$ is always a subring of S.

The kernel $\text{Ker}(f)$ is not normally a subring but satisfy:

- Closed under addition.
- Closed under multiplication (absorption property).

A subset $I \subset R$ of a ring R is an ideal if it is closed under addition, left and right absorption.

The (left) principle ideal of a ring R generated by $a \in R$ is defined as

$$(a) := \{ra \mid r \in R\}$$

and the ideal generated by elements $a_1, \cdots, a_n \in R$ is defined as

$$(a_1, \cdots, a_n) := \{r_1 a_1 + \cdots + r_n a_n \mid r_i \in R\}$$

A useful map is an evaluation map $\varphi : F[x] \to \text{Im}\varphi$ given by $\varphi(f) = f(a)$ for some $a \in F$.

6.1 Equivalence Classes Defined by an Ideal

Let R be a ring and $I \subset R$ an ideal, two elements $x, y \in R$ are equivalent if $x - y \in I$.

The cosets of $x \in R$ modulo I is the set $\bar{x} := \{x + r \mid r \in I\}$ and x is a representative of $\bar{x} = x + I$. The cosets partition the ring R.

6.2 The Quotient Ring

The quotient ring R/I where R is a ring and $I \subset R$ is an ideal, is defined as

$$R/I := \{x + I \mid x \in R\}$$

with addition and multiplication on the representatives.

The quotient $R/(f)$ is an integral domain if and only if f is irreducible over R.

6.3 The Quotient Map and The First Isomorphism Theorem for Rings

Let R be a ring, we have that $\xi : R \to R/I$ given by $\xi(r) = \bar{r}$ is an isomorphism, this is called the quotient map or the canonical map.

Every ideal $I \subset R$ is the kernel of some homomorphism $f : R \to S$ for some ring S.

The First Isomorphism Theorem for Rings states that if $\varphi : R \to S$ is a homomorphism, then the map

$$\begin{aligned} \Phi : & R/\mathrm{Ker}(\varphi) \to \mathrm{Im}(\varphi) \\ & \bar{x} \mapsto \varphi(x) \end{aligned}$$

is an isomorphism and $R/\mathrm{Ker}(\varphi) \cong \mathrm{Im}(\varphi)$.

6.4 Prime and Maximal Ideals

Let R be a commutative ring, and $I \subset R$ be a proper (i.e $I \neq R$) ideal. Then I is a prime ideal if for all $a, b \in R$ with $ab \in I$, then $a \in I$ or $b \in I$.

Furthermore, I is a maximal ideal if the only ideals of R containing I are itself or R.

If $x \in R$ is a prime element then (x) is a prime ideal.

If $I \subset R$ is a maximal ideal, then it is prime.

Let R be a commutative ring, and $I \subset R$ and ideal. We have:

1. I is a prime ideal if and only if R/I is an integral domain.
2. I is a maximal ideal if and only if R/I is a field.

7 Principal Ideal Domains

An integral domain in which all ideals are principle is a Principle Ideal Domain.

Any Principle Integral Domain is a Unique Factorisation Domain.

7.1 Fields as Quotients

Let F be a field and $f(x) \in F[x]$ be irreducible. Then $F[x]/(f(x))$ is a field and is a vector space over F with basis $B := \{1, \bar{x}, \cdots, \bar{x}^{n-1}\}$

7.2 Finite Fields

For every prime $p \in \mathbb{Z}$ and $n \in \mathbb{N}$, there exists an irreducible polynomial $f(x) \in (\mathbb{Z}/p)[x]$ of degree n. Thus $(\mathbb{Z}/p)[x]/(f(x))$ is a field with p^n elements. Any two such fields are isomorphic, and we denote the unique (up to isomorphism) field with p^n elements as $\mathbb{F}_{p^n}$.

8 The Chinese Remainder Theorem

Let R be a Principle Ideal Domain and $a, b \in R$ be coprime elements (that is no irreducible element divides both a and b). Then $(a, b) = (1)$ and $ax + by = 1$ for some $x, y \in R$.

The Chinese Remainder Theorem states that for R a Principle Ideal Domain, if $a_1, \cdots, a_k$ are pairwise coprime, then the map

$$\begin{aligned} \Xi :& R/(a_1 \cdots a_k) \to R/(a_1) \times \cdots R/(a_k) \\ & r + (a_1 \cdots a_k) \mapsto (r + (a_1), \cdots, r + (a_k)) \end{aligned}$$

is an isomorphism, that is $R/(a_1 \cdots a_k) \cong R/(a_1) \times \cdots R/(a_k)$.

9 Basics on Groups

A group G is a set together with a binary operation $\star : G \times G \to G$ such that

1. $(g \star f) \star h = g \star (f \star h)$ for all $g, f, h \in G$.
2. $\exists\, e \in G$ such that $e \star g = g$ for all $g \in G$.
3. For every $g \in G$, there exists $h \in G$ such that $g \star h = h \star g = e$.

An abelian group is a group whose operation is commutative, that is $g \star h = h \star g$ for all $g, h \in G$.

For $g, h \in G$ with G a group, $(g \star h)^{-1} = h^{-1} \star g^{-1}$.

10 The Dihedral Groups

For $n \geq 3$, the dihedral group is defined as

$$\mathrm{D}_n = \langle r, s \mid r^n = e, s^2 = e, srs = r^{-1} \rangle$$

Any element in D_n can be written uniquely as $r^a s^b$ with $a \in \{0, 1, \cdots, n-1\}$ and $b \in \{0, 1\}$, which means $|\mathrm{D}_n| = 2n$.

11 Generators, Orders, & Cyclic Groups

A set S is a set of generators of a group G if any element $g \in G$ can be written as a product of elements in S, then $G = \langle S \rangle$.

If a group is generated by a single element, then the group is cyclic.

The number of elements in a group G is the order of G. The order of an element $g \in G$ is the smallest positive integer such that $g^n = e$.

If $G = \langle x \rangle$ is finite of order n, then $\mathrm{ord}(x^a) = \frac{n}{\gcd(n,a)}$ and $\langle x \rangle = \langle x^a \rangle$ if and only if $\gcd(n, a) = 1$.

12 The Symmetric Groups

A permutation of $S = \{1, \cdots, n\}$ is a bijection $\sigma : S \to S$ often written as

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}$$

A k-cycle is a permutation such that for some $1 \le k \le n$ we have

$$\sigma(a_1) = a_2,\ \sigma(a_2) = a_3,\ \cdots,\ \sigma(a_k) = a_1$$

and that $\sigma(a) = a$ for all $a \neq a_i$; this is denoted by $(a_1\ a_2\ \cdots\ a_k)$. A transposition is a cycle of length 2.

For $n \geq 1$, the symmetric group is the group of all permutations of $S = \{1, \cdots, n\}$, denoted as S_n. We have that $|\mathrm{S}_n| = n!$.

Two cycles are disjoint cycles if they do not both permute the same element.

We have the following:

1. Disjoint cycles commute with each other.
2. Every permutation is a product of disjoint cycles, and this is unique up to the order of the cycles and the different ways to write a cycle.

Every permutation is a product of transpositions (but not uniquely), and thus S_n is generated by transpositions.

For a σ a k-cycle, $\mathrm{ord}(\sigma) = k$. If $\sigma = \sigma_1 \cdots \sigma_n$ where each σ_i is a k_i-cycle and are all pairwise disjoint, then $\mathrm{ord}(\sigma) = \mathrm{lcm}(k_1, \cdots, k_n)$.

The inverse of a k-cycle is the cycle written backwards, that is if $\sigma = (a_1 \ \cdots \ a_k)$, then $\sigma^{-1} = (a_k \ \cdots a_1)$.

Two elements $g, g' \in G$ are conjugate in G if $hgh^{-1} = g'$ for some $h \in G$.

If $\sigma = (a_1 \ \cdots \ a_k) \in \mathrm{S}_n$ is a k-cycle and $\lambda \in \mathrm{S}_n$, then

$$\lambda\sigma\lambda^{-1} = (\lambda(a_1) \ \cdots \ \lambda(a_k))$$

and a conjugate of a cycle in S_n is always a cycle of the same length.

13 Subgroups & Cosets

Give a group G, a subset $H \subseteq G$ is a subgroup if:

1. $e \in H$.
2. For $h_1, h_2 \in H$, $h_1 h_2 \in H$.
3. For $h \in H$, $h^{-1} \in H$.

A subgroup $H \subseteq G$ is proper if $H \neq G$ and trivial if $H = \{e\}$.

Lagrange's Theorem states that for a finite group G, and a subgroup $H \subseteq G$, the order of the subgroup $|H|$ divides the order of the group $|G|$.

Let $g \in G$, the left/right coset of H with respect to g is

$$gH := \{gh \mid h \in H\} \quad \text{or} \quad Hg := \{hg \mid h \in H\}$$

which coincide when G is abelian. The cosets form a disjoint partition of the group G, that is $G = \bigsqcup_{g \in G} gH$.

If $xH = yH$, then $y^{-1}x \in H$.

Let G be a finite group and $H \subseteq G$ a subgroup. Then every (left) coset of H in G has precisely $|H|$ elements. In particular, $|G| = m \cdot |H|$, where m is the number of (left) cosets.

Let G be a finite group with $g \in G$, then $\mathrm{ord}(g)$ divides $|G|$.

Let G be a group and $H \subseteq G$ a subgroup, the index of H in G, denoted $[G : H]$, is the number of cosets in G or if G is finite, then $[G : H] = |G|/|H|$.

14 Homomorphisms & Isomorphisms

Let G and H be groups, a group homomorphism $\varphi : G \to H$ is a map such that for all $x, y \in G$

$$\varphi(xy) = \varphi(x)\varphi(y)$$

An group isomorphism is a bijective homomorphism, if we have an isomorphism between two groups, then they are isomorphism and $G \cong H$.

For G a group and $H \subseteq G$ is normal if for every $g \in G$ and $h \in H$, we have $g^{-1}hg \in H$, denoted by $H \triangleleft G$.

For G a group and N a normal subgroup, the quotient group G/N, is the group defined as

$$G/N := \{gN \mid g \in G\}$$

with $|G/N| = [G : N]$.

Let $\varphi : G \to H$ be a group homomorphism, then $\mathrm{Ker}(\varphi)$ is normal in G. Conversely, if N is a normal subgroup of G, then N is the kernel of a group homomorphism.

Let $\varphi : G \to H$ be a group homomorphism. The First Isomorphism Theorem for Groups states that $G/\mathrm{Ker}(\varphi) \cong \mathrm{Im}(\varphi)$.

15 Relating & Identifying Finite Groups

For G a cyclic group of order n, then $G \cong \mathbb{Z}/n$.

For G a group of order p, then G is cyclic and $G \cong \mathbb{Z}/p$.

We have that $\mathrm{D}_3 \cong \mathrm{S}_3$.

If G and H are two groups, their direct product is $G \times H := \{(g, h) \mid g \in G, h \in H\}$ with operations pairwise.

The Klein 4-group is given by $\mathrm{K}_4 = \mathbb{Z}/2 \times \mathbb{Z}/2$.

We define quaternions Q_8 as the group,

$$\mathrm{Q}_8 := \langle -1, i, j, k \mid (-1)^2 = 1, i^2 = j^2 = k^2 = ijk = -1 \rangle$$

Order	Groups
1	$\{e\}$
2	$\mathbb{Z}/2$
3	$\mathbb{Z}/3$
4	$\mathbb{Z}/4,\ \mathrm{K}_4$
5	$\mathbb{Z}/5$
6	$\mathbb{Z}/6,\ \mathrm{S}_3 \cong \mathrm{D}_3$
7	$\mathbb{Z}/7$
8	$\mathbb{Z}/8,\ \mathbb{Z}/2 \times \mathbb{Z}/4,\ (\mathbb{Z}/2)^3,\ \mathrm{D}_4,\ \mathrm{Q}_8$

Groups of small order up to isomorphism.

16 The Alternating Groups

Let $\sigma \in \mathrm{S}_n$. A pair (i, j) is an inversion of σ if $\sigma(i) > \sigma(j)$.

The sign of a permutation σ is defined by $\mathrm{sgn}(\sigma) = (-1)^N$. A permutation is even if we have $\mathrm{sgn}(\sigma) = 1$ and odd if $\mathrm{sgn}(\sigma) = -1$.

Let $\sigma \in \mathrm{S}_n$ be a permutation, and $\tau \in \mathrm{S}_n$ a transposition, then $\mathrm{sgn}(\sigma\tau) = -\mathrm{sgn}(\sigma)$.

A permutation $\sigma \in \mathrm{S}_n$ is odd if the number of transpositions in its factorisation is odd, and σ is even otherwise. In particular, the parity of the number of transpositions needed in any factorisation of $\sigma \in \mathrm{S}_n$ is always the same.

Let C_2 be the group of order 2 written in multiplicative form, then $\mathrm{sgn} : \mathrm{S}_n \to \mathrm{C}_2$ is a group homomorphism.

The alternating group A_n is the subgroup of S_n consisting of even permutations. We have the following.

1. $|\mathrm{A}_n| = n!/2$.
2. $\mathrm{A}_n \triangleleft \mathrm{S}_n$.
3. The group A_n is generated by 3-cycles.

17 Linear Groups

Let R be a commutative ring and $n \in \mathbb{N}$. The general linear group over R is,

$$\mathrm{GL}_n(R) := \{A \in \mathrm{M}_n(R) \mid \det(A) \in R^\times\}$$

with the special linear group over R is,

$$\mathrm{SL}_n(R) := \{A \in \mathrm{M}_n(R) \mid \det(A) = 1\}$$

We have that $\mathrm{GL}_n(R)/\mathrm{SL}_n(R) \cong R^\times$.

Let $\mathbb{F}_q$ be the finite field with q elements,

$$|\mathrm{GL}_n(\mathbb{F}_q)| = (q^n - 1)\cdots(q^n - q^{n-1})$$

and also that

$$|\mathrm{SL}_n(\mathbb{F}_q)| = (q^n - 1)\cdots(q^n - q^{n-1})/(q-1)$$

18 Group Actions

The following describes a group action.

For a group G and set X, the action of G on X is a map,

$$\begin{aligned} * : G \times X &\to X \\ (g, x) &\longmapsto g * x \end{aligned}$$

which often abbreviated to gx satisfying:

1. $g * (h * x) = (gh) * x$ for all $g, h \in G$ and $x \in X$.
2. $e * x = x$ for all $x \in X$.

If we take a subgroup $H \subseteq G$, then H acts on X via the same action as G; this is called the induced action of H on X.

Given a set X, the set S_X is the set of all bijections from X to itself; this is a group. Moreover, G acts on X if there is a group homomorphism from G to S_X.

The following is Cayley's Theorem.

Every finite group G is isomorphic to a subgroup of some symmetric group.

18.1 Finding the Symmetric Group Representation of a Finite Group

The following is the method to find the symmetric group representation of a finite group.

1. Enumerate the elements of G.

2. Consider G acting on itself under left multiplication, that is $\varphi_g : G \to G$ where $g, x \in G$ and $\varphi_g(x) = gx$
3. For each element $g \in G$ find $\varphi_g(x)$ for all $x \in G$.
4. Match up each element as a cycle.

If G has some specific structure, find the elements who match to the generators.

18.2 Orbits and Stabilisers

Let G act on X. For $x \in X$, we have the orbit of $x \in X$ as

$$\mathrm{Orb}(x) = \{gx \mid g \in G\} \subset X$$

and the stabiliser of $x \in X$ in G as

$$\mathrm{Stab}(x) = \{g \in G \mid gx = x\} \subset G$$

The stabiliser of $x \in X$ is a subgroup of G and the orbits of $x \in X$ disjointly partition X.

18.3 Cosets and Conjugacy Classes via Orbits

Consider a group G acting on itself by left multiplication, that is $x \mapsto gx$ for all $g, x \in G$. Take a subgroup $H \subseteq G$, and take the induced action of H on G, for $x \in G$,

$$\mathrm{Orb}(x) = \{hx \mid h \in H\} = Hx$$

and similarly, we can obtain the right cosets by action under right multiplication.

Similarly, consider a group G acting on itself by conjugation, that is $x \mapsto gxg^{-1}$. We have that under conjugation, for $x \in G$,

$$\mathrm{Orb}(x) = \{gxg^{-1} \mid g \in G\}$$

which is the conjugacy class of x, and further

$$\mathrm{Stab}(x) = \{g \in G \mid gxg^{-1} = x\}$$

which is the centraliser of x, usually denoted $C_G(x)$.

18.4 The Orbit-Stabiliser Theorem

The following is The Orbit-Stabiliser Theorem. For a finite G, if G acts on X then we have that $|\mathrm{Orb}(x)| = |G|/|\mathrm{Stab}(x)|$ for every $x \in X$.

18.5 Transitive Actions

An action of G on X is transitive if $\mathrm{Orb}(x) = X$ for every $x \in X$.

Under a transitive action, the orbits are the same, that is $\mathrm{Orb}(x) = \mathrm{Orb}(y)$ for all $x, y \in X$. Moreover, there exists a $g \in G$ such that $\mathrm{Stab}(y) = g\mathrm{Stab}(x)g^{-1}$.

19 Conjugacy in Symmetric Groups

Let $\sigma \in S_n$ be the product of disjoint cycles, that is $\sigma = \sigma_1 \cdots \sigma_m$ where σ_i has length n_i and $n_1 \leq \cdots \leq n_m$ (including 1-cycles). The cycle type of σ is the list $[n_1, \cdots, n_m]$.

Two elements in S_n are conjugate if and only if they have the same cycle type.

The conjugacy classes in S_n are given by sets of elements of the same cycle type. Each cycle type, and hence each conjugacy class, is determined by a partition of n.

19.1 Finding the Conjugacy Classes of S_n

The following is the method to find the conjugacy classes of S_n.

1. Find all partitions of n.
2. Each partition $[n_1, \cdots, n_k]$ is a class with permutations corresponding to the length, for example $[1, 2, 2, 3]$ represents a class of permutations of the form
$$(ab)(cd)(efg)$$
3. To find the size of each class, preform some counting, for example for the class $[1, 2, 2, 3]$ would have a size of
$$\frac{\left(\frac{8\times 7}{2}\right)\left(\frac{6\times 5}{2}\right)}{2} \times \left(\frac{4 \times 3 \times 2}{3}\right) = 1680$$

19.2 Normal Subgroups of S_n via Conjugacy Classes

A normal subgroup is a union of conjugacy classes. The following is the method to find the normal subgroups of S_n.

1. Find the conjugacy classes and their size.
2. Use Lagrange's Theorem to find the possible combinations of conjugacy classes to make a subgroup.
3. Check if this is in fact a subgroup, if so it is normal.

20 Cauchy's Theorem and Groups of Order $2p$

The following is Cauchy's Theorem.

Let G be a finite group and p-prime such that $p \,\big|\, |G|$, then G has a cyclic subgroup of order p.

The following is Sylow's Theorem which is a more general version of Cauchy's Theorem.

Let G be a finite group and p-prime such that $|G| = p^n m$ where $\gcd(p, m) = 1$, then G has a subgroup of order p^n.

The subgroup in Sylow's Theorem is called a Sylow p-subgroup.

We have the following as a consequence of Cauchy's Theorem.

Let G be a finite group of order $2p$ for p-prime different from 2, then $G \cong \mathbb{Z}/(2p)$ or D_p.

Groups whose order is a power of a prime are called p-groups.

21 Finite Simple Groups

A group G is simple if it has no non-proper normal subgroups.

We have the following.

- An abelian group is simple if and only if it is isomorphic to $\mathbb{Z}/p$ for p—-prime.
- S_n for $n \geq 3$ is not simple.
- A_n is simple for $n \geq 5$.

22 Abelian Groups

22.1 Finite Abelian Groups

The following is known as The Fundamental Theorem for Finite Abelian Groups.

Let G be a finite and abelian group, then

$$G \cong \mathbb{Z}/a_1 \times \cdots \times \mathbb{Z}/a_n$$

for some a_i's such that $a_1 \geq 2$ and $a_1 \mid a_2 \mid \cdots \mid a_n$ - this representation is unique. Moreover, if G has order k then we require that $a_1 \cdots a_n = k$.

The following is The Chinese Remainder Theorem.

Let $m, n \in \mathbb{Z}$ with $\gcd(m, n) = 1$, then

$$\mathbb{Z}/mn \cong \mathbb{Z}/n \times \mathbb{Z}/m$$

22.2 Finding All Abelian Groups of Order k

We wish to find all abelian groups (up to isomorphisms) of order k.

1. Factorise k as $k = p_1^{\alpha_1} \cdots p_j^{\alpha_j}$ for primes p_j and powers α_j.
2. Find $\{a_i\}_{i=1}^n$ such that
   (a) $a_1 \mid \cdots \mid a_n$
   (b) $a_1 \cdots a_n = k$

   for each $1 \leq n \leq k$.

22.3 Finitely Generated Abelian Groups

A group G is finitely generated if it's generating set $S \subset G$ is finite.

Given a finitely generated abelian group G, we have that $G \cong \mathbb{Z}^n/K$ for some subgroup $K \subset \mathbb{Z}^n$. The subgroup K is called the relation subgroup of G and its elements are called relations.

If the relation subgroup of G is $K = \{e\}$, then G is called a free abelian group of rank n.

The following is The Fundamental Theorem of Finitely Generated Abelian Groups.

Let G be a finitely generated abelian group, then

$$G \cong \mathbb{Z}/a_1 \times \cdots \mathbb{Z}/a_n \times \mathbb{Z}^r$$

for $a_i \geq 1$ and $a_1 \mid a_2 \mid \cdots \mid a_n$ - this representation is unique. The a_i's are the torsion coefficients and r is the rank of G.

Given a group with generators, we can look that the relation matrix in order to find the torsion coefficients and the rank.

The following is an example. Let G be generated by the elements x, y, z with relations $x^6y^{-3}z^9 = e$ and $x^2y^{-8}z^4 = e$. The *relation matrix* is given by

$$A = \begin{pmatrix} 6 & -3 & 9 \\ 2 & -8 & 4 \end{pmatrix}$$

we then use row and column operations to put the matrix in the following form.

$$\begin{pmatrix} 6 & -3 & 9 \\ 2 & -8 & 4 \end{pmatrix} \underset{\sim}{\scriptstyle r_1 \leftrightarrow r_2} \begin{pmatrix} 2 & -8 & 4 \\ 6 & -3 & 9 \end{pmatrix} \underset{\sim}{\scriptstyle r_2 \to 2r_1 - r_2} \begin{pmatrix} 2 & -8 & 4 \\ 0 & -21 & 3 \end{pmatrix} \underset{\sim}{\scriptstyle c_2 \to 4c_1 + c_2} \begin{pmatrix} 2 & 0 & 4 \\ 0 & -21 & 3 \end{pmatrix}$$
$$\underset{\sim}{\scriptstyle c_3 \leftrightarrow c_2} \begin{pmatrix} 2 & 0 & 0 \\ 0 & -3 & -21 \end{pmatrix} \underset{\sim}{\scriptstyle c_3 \to 7c_2 - c_3} \begin{pmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \end{pmatrix} \underset{\sim}{\scriptstyle c_2 \to -c_2} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$$

The diagonal entries are then the torsion coefficients, with the number of generators subtracted by the number of torsion coefficients as the rank. That is,

$$G \cong \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z} \cong \mathbb{Z}/6 \times \mathbb{Z}$$

where we used the Chinese Remainder Theorem to get the coefficients in the right form.

In the above example, we reduced A into In the above example, we reduced A into Smith normal form. A $n \times m$ matrix A is in Smith normal form if it can we written as

$$\operatorname{diag}(\alpha_1, \cdots, \alpha_k, 0 \cdots, 0)$$

for some α_i's using the following operations.

Operation 1. Adding a multiple of a row to another row.

Operation 2. Adding a multiple of a column to another column.

Operation 3. Multiply a row or column by ± 1.

22.4 Finding the Representation of a Finitely Generated Abelian Group

We have the following method for finding the representation of a finitely generated abelian group with relation matrix K.

Given a finitely generated abelian group G with $K = \langle g_1, \cdots, g_n \mid \text{relations} \rangle$, we proceed by the following.

1. Write down the relation matrix A.
2. Reduce A into,
$$\operatorname{diag}(d_1, \cdots, d_m, 0, \cdots, 0)$$
with $d_1 \mid d_2 \mid \cdots \mid d_m$. That is to say into Smith normal form with an extra condition.
3. The torsion coefficients are then $d_1, \cdots, d_m$ with the rank being $n - \operatorname{rank}(A)$ where often this will be the number of rows of A.